

ECE4010J Probabilistic Methods in Engineering (Summer 2026)

Assignment 5

Date Due: See canvas

This assignment has a total of **(150 points)**.

Note: Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

Exercise 5.1 (50 pts) Let rv (X, Y) follow the bivariate normal distribution with pdf

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}$$

and parameters $\sigma_X, \sigma_Y > 0$, $\mu_X, \mu_Y \in \mathbb{R}$, and $|\rho| < 1$.

- (i) (10 pts) Verify that the marginal density for X is that of a normal distribution with mean μ_X and variance σ_X^2 .
- (ii) (10 pts) Verify that ρ is the coefficient of correlation between X and Y .
- (iii) (10 pts) Show that X and Y are independent iff $\rho = 0$.
- (iv) (10 pts) Show that

$$\mathbb{E}[Y | X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X)$$

- (v) (10 pts) The life X of a tube and the lument diameter Y are distributed as a bivariate normal random variable with the parameters $\mu_X = 2000$ hours, $\mu_Y = 0.1$ inch, $\sigma_X^2 = 2500$ hours², $\sigma_Y^2 = 0.01$ inch², and $\rho = 0.87$.

The quality-control manager wishes to determine the life of each tube by measuring the lument diameter. If a lument diameter is 0.098 inch, what is the probability that the tube will last 1950 hours or longer?

Exercise 5.2 (50 pts) Let

$$f_{XYZ}(x, y, z) = \frac{3z^2}{7\sqrt{2\pi}} e^{-zy} \exp\left[-\frac{1}{2}\left(\frac{x-y}{z}\right)^2\right] \quad (1)$$

for $y \geq 0$ and $1 \leq z \leq 2$, and $f_{XYZ}(x, y, z) = 0$ otherwise. Find

- (i) (5 pts) Verify that f_{XYZ} is a density;
- (ii) (10 pts) $f_{YZ}(y, z)$ and $f_{X|YZ}(x | y, z)$;
- (iii) (15 pts) $f_Z(z)$, $f_{Y|Z}(y | z)$, and $f_{XY|Z}(x, y | z)$;
- (iv) (20 pts) $\mathbb{E}[X]$, $\mathbb{E}[XZ]$, $\mathbb{E}[XY]$, $\mathbb{E}[YZ]$, and $\mathbb{E}[XYZ]$.

Exercise 5.3 (10 pts) [PR12] Suppose the random points $X = (X_1, X_2, X_3) \in \mathbb{R}^3$ are distributed uniformly on the unit sphere $S^2 = \{x = (x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1^2 + x_2^2 + x_3^2 = 1\}$, as shown in Fig. 1. What is the pdf of X_1 ?

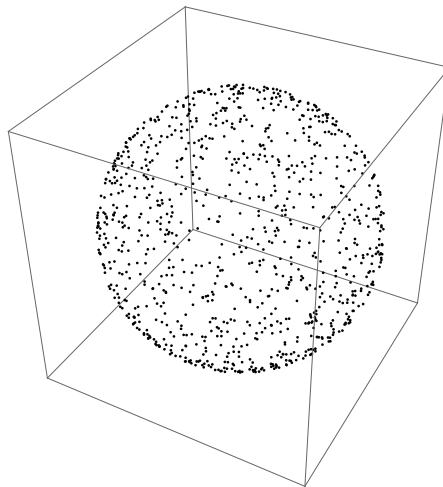


Figure 1: Uniform Spherical Distribution

Exercise 5.4 (10 pts) Given rv's N and X . Suppose the rv $N \sim \text{Poisson}(\lambda)$, and $X \mid (N = n) \sim \text{Binomial}(n, p)$. Find the distribution of X (in terms of pmf or pdf).

Exercise 5.5 (20 pts) Verification of Huygens' Principle [Fri12]. A point source is located at S in Fig. 2. It radiates uniformly in all directions, within the plane of the figure, so that the probability density $p_\Theta(\theta)$ for direction Θ of a photon follows a uniform distribution

$$p_\Theta(\theta) = \begin{cases} 1/(2\pi), & 0 \leq \theta \leq 2\pi \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

- (i) (10 pts) Show that the probability density $p_X(x)$ for photons crossing the line AB located distance a from S obeys a Cauchy law with parameter a (after suitable renormalization, since half the photons never cross AB). The pdf for Cauchy (or Lorentz) law with parameter a is given by

$$p(x) = \frac{1}{\pi} \frac{a}{a^2 + x^2} \quad (3)$$

- (ii) (10 pts) Suppose the source S is removed, replaced with a line source along AB whose intensity/length obeys the previously found $p_X(x)$. Each point S' on the line source radiates uniformly in all directions, as did S . Let line $A'B'$ be distance b from the former source S . Show that the intensity $p_Y(y)$ along $A'B'$ is exactly Cauchy with parameter b , as if there were instead a real point source at S once again.

hint:

$$p_Y(y) = \int_{-\infty}^{\infty} p(y \mid x) p_X(x) dx \quad (4)$$

This verifies Huygens' principle, which states that each point on the wavefront of a propagating light wave acts as a new source of secondary light waves, from which the light wave at later times may be constructed.

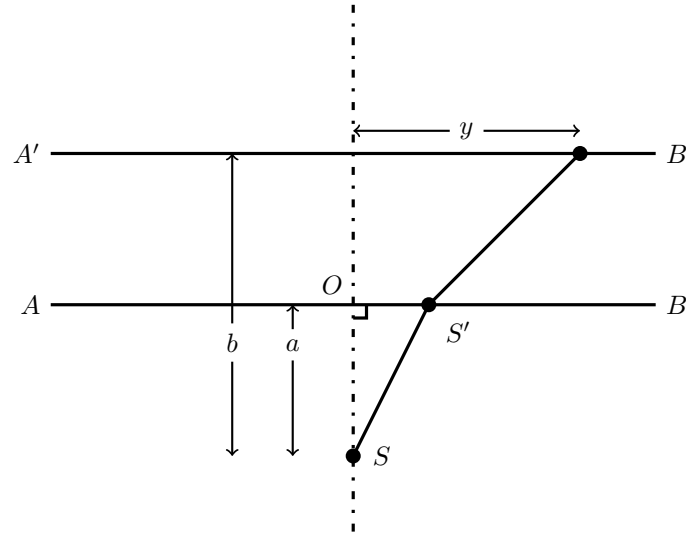


Figure 2: Huygens' principle from properties of the Cauchy probability law. The light distribution along $A'B'$ due to the point source S is attained alternatively by a line source along AB with the proper intensity profile.

Exercise 5.6 (10 pts) Given a real symmetric *positive definite*¹ matrix $A \in M_n(\mathbb{R})$, vector $v \in \mathbb{R}^n$, and another real symmetric matrix $D \in M_n(\mathbb{R})$, show that

$$\int_{\mathbb{R}^n} e^{-\frac{1}{2}x^\top Ax + v^\top x} (x^\top D x) dx = [v^\top A^{-1} D A^{-1} v + \text{tr}(D A^{-1})] \frac{(2\pi)^{n/2}}{\sqrt{\det A}} e^{\frac{1}{2}v^\top A^{-1} v}$$

hint: Consider a random vector $X \sim \text{Normal}(A^{-1}v, A^{-1})$.

References

- [Fri12] B.R. Frieden. *Probability, Statistical Optics, and Data Testing: A Problem Solving Approach*. Springer Series in Information Sciences. Springer Berlin Heidelberg, 2012 (Cited on page 2).
- [PR12] Jim Pitman and Nathan Ross. "Archimedes, Gauss, and Stein". In: *Notices AMS* 59 (2012), pp. 1416–1421 (Cited on page 1).

¹ A is positive definite if $v^\top A v > 0$ for all $v \in \mathbb{R}^n \setminus \{0\}$.